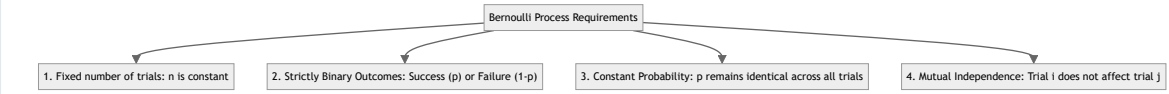
Lecture 20.18: Binomial Distribution

📌 Topic Overview

In this lecture, Shraddha ma'am details the **Binomial Distribution**, establishing the 4 strict Bernoulli criteria, deriving the combination formula on the blackboard, and solving the classic 5-coin toss problem ($P(3 \text{ Heads in } 5 \text{ tosses}) = \frac{5}{16} = 0.3125$) using both manual mathematics and `scipy.stats.binom`.

🏛️ The 4 Bernoulli Trial Conditions



📐 Mathematical Formulation

Let random variable $X \sim \text{Binomial}(n, p)$ count the number of successes:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

Parameters:

- **Mean (Expected Value):** $\mu = n \cdot p$
- **Variance:** $\sigma^2 = n \cdot p \cdot (1-p)$
- **Standard Deviation:** $\sigma = \sqrt{n \cdot p \cdot (1-p)}$

Blackboard Example: 5 Coin Tosses, Exactly 3 Heads

- $n = 5, k = 3, p = 0.5, 1-p = 0.5$

$$P(X = 3) = \binom{5}{3} (0.5)^3 (0.5)^2 = \frac{5!}{3!2!} \left(\frac{1}{8}\right) \left(\frac{1}{4}\right) = 10 \times \frac{1}{32} = \frac{5}{16} = 0.3125 \quad (31.$$

💻 Python Implementation

```
from scipy.stats import binom

def compute_binomial_probability(n=5, k=3, p=0.5):
    """
    Computes exact Binomial probability P(X = k) for k successes in n Bernoulli tr
    Demonstrates analytical combination formula and scipy.stats.binom verification

    Parameters:
        n (int): Number of independent trials.
        k (int): Number of successful outcomes.
        p (float): Success probability on a single trial.

    Returns:
        dict: Mean, Variance, analytical and scipy PMF probabilities.
    """
    # Mean and Variance formulas: mu = n*p, var = n*p*(1-p)
    mean_val = n * p
    variance_val = n * p * (1 - p)

    # Scipy implementation
    prob_k = float(binom.pmf(k, n, p))

    return {
        "n_trials": n,
        "k_successes": k,
        "p_single": p,
        "mean_np": mean_val,
        "variance_npq": variance_val,
        "P(X=k)": prob_k,
        "fraction_str": "10 * (1/32) = 5/16 = 0.3125"
    }

# --- Execution & Verification ---
# Shraddha ma'am's coin problem: Toss 5 times, find probability of exactly 3 heads
res = compute_binomial_probability(n=5, k=3, p=0.5)
print(f"Trials n = {res['n_trials']}, Heads k = {res['k_successes']}, p = {res['p_
print(f"Theoretical Probability: {res['fraction_str']}")
print(f"Scipy binom.pmf result : {res['P(X=k)']:.4f}")
print(f"Mean np = {res['mean_np']}, Variance np(1-p) = {res['variance_npq']:.2f}")
```

🔑 Key Takeaways

1. A Binomial distribution is simply the sum of n independent Bernoulli trials.
2. $\binom{n}{k}$ accounts for all the different ordering permutations in which k successes can occur among n trials.
3. In deep learning and NLP, A/B testing conversion rates and click-through rates are modeled using Binomial distributions.